

Aditya Shukla

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EDUCATION

VIT Bhopal University

B.Tech in Computer Science (Specialization in AI/ML)

Bhopal, M.P.

Sept 2023 - May 2027

Rani Laxmi Bai Memorial School

Senior Secondary

Lucknow, U.P.

EXPERIENCE

Machine Learning Intern

Bodhganga Academy

Apr 2026 – Jul 2026

Delhi (Hybrid)

- Developed ML models for personalized learning recommendations using user behavior and educational data.
- Built data preprocessing and feature engineering pipelines for efficient model training and evaluation.
- Integrated ML inference APIs with the web platform and optimized prediction performance through model monitoring.

PROJECTS

Fake News Detection System | *Python, Flask, Scikit-Learn, Pandas, NumPy, HTML, CSS* |

- Developed a web app to classify news articles as real, fake, or unreliable using Logistic Regression, Decision Tree and Gradient Boosting
- Built Flask backend for model inference and API handling, implemented ensemble-style logic to improve classification reliability
- Integrated NewsAPI to fetch real-time news articles and provide AI-powered predictions through a responsive Flask web interface.

Lane Detection & Driver Assistance System | *Python, OpenCV, NumPy, Computer Vision* |

- Developed a real-time lane detection system using OpenCV and NumPy for processing image and video streams.
- Implemented edge detection, HLS color masking, and Region of Interest (ROI) filtering to accurately identify lane boundaries under varying road conditions.
- Integrated Optical Flow and Harris Corner Detection techniques to improve lane tracking stability and motion estimation across consecutive frames.
- Designed a lane departure warning system with confidence scoring and real-time visual overlays to enhance driving assistance.

Helmet Detection using YOLOv8 | *Python, YOLOv8, OpenCV, Computer Vision* |

- Developed an object detection system using YOLOv8 to identify motorcyclists with and without helmets from images and video streams.
- Converted PASCAL VOC annotations to YOLO format and organized train, validation, and test datasets for efficient model training.
- Fine-tuned a pretrained YOLOv8n model on a custom dataset and implemented an inference pipeline generating bounding boxes, class labels, and confidence scores.
- Evaluated model performance using detection metrics and analyzed false positives and false negatives to identify dataset and training limitations.

TECHNICAL SKILLS

Languages: Python, C++, SQL

Libraries & Frameworks: Flask, Scikit-learn, OpenCV, NumPy, Pandas, Ultralytics YOLO

Web: HTML, CSS, JavaScript

Tools: Git, GitHub, VS Code, Postman, AWS

APIs: REST APIs, News API

Concepts: Machine Learning, Computer Vision, Feature Engineering, Model Evaluation

CERTIFICATIONS

Applied Machine Learning in Python

University of Michigan (Coursera)

Cloud Computing

NPTEL (IIT Kharagpur)

Introduction to Internet of Things

NPTEL (IIT Kharagpur)

Student Innovation Workshop

T-Hub, Hyderabad